When Reputation Backfires: Gossip and Social Pressure Undermine Contribution in LLM Agent Societies under Climate Risk

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Abstract

Reputation systems are widely treated as stabilizers of cooperation in multi-agent societies. We study whether that assumption holds for large language model (LLM) agents in a climate-risk public-goods economy. Twenty-six agents interact for thirty rounds under forced institutional assignment, climatic shocks, a dual-use loss-and-damage pool, democratic rule updating, and social-information channels that include peer consistency scoring, gossip bulletins, and aggregate reputation. The primary outcome is contribution intensity relative to wealth. After bad-reputation and gossip events, mean next-round contribution intensity declines. The decline is largest for agents in the non-enforcing institution after gossip targeting. Coded rationales after these events show opportunistic framing far more often than reputation repair. Average cooperation remains moderately positive, yet path stability is limited and norm emergence is mixed. Boundary-condition evidence from shock-free free-choice baselines suggests reputation can coordinate institutional sorting when exit is available. Under climate stress and forced assignment, social pressure associates with withdrawal. The finding challenges the default claim that social monitoring raises cooperation in LLM agent societies.

Keywords: large language model agents; multi-agent systems; reputation; gossip; cooperation; climate risk sharing; endogenous institutions; public goods

1. Introduction

Cooperation among artificial agents is often assumed to improve when peers can observe and discuss one another. Reputation, image scoring, and gossip are classic mechanisms in that tradition (Wedekind and Milinski 2000; Milinski et al. 2002; Nowak 2006). Recent work transplants these mechanisms into societies of large language model agents (Park et al. 2023; Piatti et al. 2024; Ren et al. 2025a). The shared premise is simple. Social visibility should raise contributions by making free-riding costly in status.

We report a contrary pattern. In an LLM multi-agent climate-risk economy, gossip and bad-reputation events are followed by lower contribution intensity on average. Agents rarely narrate repair. They more often justify reduced giving with payoff maximization and free-riding language. The nugget is therefore precise. Under climate shocks, forced institutional assignment, and democratic adaptation, social pressure associates with withdrawal rather than image repair.

1.1 Goal, problem, and solution

The goal is to test whether social-information mechanisms that are expected to support cooperation do so for LLM agents facing climate risk sharing. The problem is that most LLM multi-agent studies either omit climate asymmetry or treat reputation as an unambiguous stabilizer. Exit options and enforcement rights also vary little across designs. The solution is a controlled thirty-round society with dual institutions, dual-use deposits into a loss-and-damage fund, deterministic shocks, endogenous rule votes, and reconstructed social events tied to contribution intensity.

The setting matters for mechanism design. Climate finance debates emphasize both mitigation contributions and compensation for loss and damage (Ostrom 1990; Hardin 1968). Computational societies can stress-test institutional packages before policy claims are made. They cannot replace empirical evaluation of real funds. They can reveal how prompted agents respond to social monitoring when capacity and vulnerability differ.

1.2 Contributions

This paper makes three contributions. First, it provides an event-study of contribution intensity after bad reputation, reputation drops, and gossip targeting in an LLM agent society. Second, it links quantitative declines to qualitative motifs that favor opportunism over repair. Third, it situates the result against boundary conditions from shock-free free-choice baselines, where reputation can anchor institutional coordination when sorting is allowed.

We do not claim a causal institution-type effect under forced assignment. Developed agents are routed to the enforcing institution. Developing agents are routed to the non-enforcing institution. Group and institution are collinear by design. Comparisons across institutions are descriptive of that joint assign-

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ment.

1.3 Roadmap

Section 2 synthesizes related work on reputation, LLM agents, and climate collective action. Section 3 defines the economic environment and information channels. Section 4 details the experimental protocol. Section 5 presents results, beginning with the social-event study. Section 6 discusses mechanisms and boundary conditions. Sections 7 and 8 state limitations and conclude. An extensive appendix documents parameters, prompt templates, and supplementary analyses.

2. Literature Review

2.1 Collective action and reputation in human societies

Collective-action problems arise when private incentives erode shared returns (Olson 1965; Hardin 1968). Laboratory and field traditions show that monitoring, punishment, and reputation can support cooperation (Fehr and Gächter 2002; Boyd et al. 2003; Ostrom 1990). Indirect reciprocity and image scoring formalize how observed giving affects future partners (Wedekind and Milinski 2000; Panchanathan and Boyd 2004). Milinski and colleagues argue that reputation helps solve commons dilemmas by linking contribution to social standing (Milinski et al. 2002). Nowak summarizes reputation among the core evolutionary routes to cooperation (Nowak 2006). Across these lines, the working hypothesis is directional. More social visibility should raise, or at least protect, cooperative transfers.

Institutional design complements reputation. Ostrom documents polycentric rules, graduated sanctions, and participatory governance in durable commons regimes (Ostrom 1990). Axelrod emphasizes reciprocity and the shadow of the future in repeated interaction (Axelrod and Hamilton 1981; Axelrod 1984). These literatures motivate dual channels in our environment. Agents face formal enforcement inside one institution and social evaluation everywhere. They also periodically revise a bounded set of fiscal and fund parameters by vote.

2.2 From classical multi-agent systems to language-model societies

Multi-agent research has long studied social dilemmas with reinforcement learners and rule-based agents (Wang and Goel 2022). Language-model agents change the measurement surface. They produce natural-language rationales, belief updates, and peer judgments in the same loop as numeric choices (Park et al. 2023; Sumers et al. 2024). Park and colleagues show that generative agents can sustain believable social routines when memory and reflection are scaffolded (Park et al. 2023). That realism invites stronger claims about social mechanisms. It also invites failure modes that numeric agents never verbalize.

Cooperation benchmarks for LLM agents report heterogeneous success across games and prompts (Madmoun and Lahlou 2025a; for Computational Linguistics 2025; Guzman Piedrahita et al. 2025). Some societies sustain contributions when communication is rich. Others collapse into free-riding once incentives dominate. Short-horizon reasoning appears repeatedly as a friction (for Computational Linguistics 2026; Ashery et al. 2025a). These findings caution against treating LLM cooperation as an automatic property of scale. They motivate designs that separate average transfer levels from norm internalization.

2.3 Reputation and communication as assumed stabilizers

Several recent systems treat reputation networks or peer talk as cooperation technology. RepuNet-style architectures encode standing so agents can condition on peer history (Ren et al. 2025a). GovSim studies sustainable cooperation among LLM agents and highlights communication and universalization reasoning as supportive channels (Piatti et al. 2024). Related communication protocols emphasize information sharing as a route to coordination (Ashery et al. 2025b; Smit and Santos 2024). The cumulative message is optimistic. If agents can talk and score one another, collective outcomes should improve.

Norm-focused work complicates that optimism without overturning it. Studies of generative-agent norms report partial emergence, cultural transmission, and fragile stability (Ren et al. 2025b; Gupta et al. 2025). Punishment-strategy transplants from Boyd and Richerson show that explicit sanctions can organize behavior when costs and targets are clear (Warnakulasuriya et al. 2025; Cross et al. 2025). Where gossip substitutes for formal punishment, the mapping is less settled. Our setting isolates that substitution for agents who cannot exit their assigned institution.

A further strand documents self-reinforcing salience loops in LLM inference (Drake 2025). Once a payoff-maximizing frame becomes salient, subsequent reasoning may recycle it. That mechanism is useful for interpreting why social stigma need not trigger repair language. It does not by itself predict the sign of contribution change. It predicts stickiness of opportunistic narratives once they appear.

2.4 Climate risk, institutions, and computational governance

Climate governance couples mitigation public goods with asymmetric vulnerability and compensation claims (Hardin 1968; Olson 1965; Ostrom 1990). Loss-and-damage instruments are dual-use in spirit. Contributions can fund both shared abatement returns and contingent payouts after shocks. Computational societies that combine those fiscal rules with LLM agents that gossip and vote remain rare.

Institutional design for multi-agent LLM systems increasingly treats incentives and sequential public-goods structure as first-class objects (Liang et al. 2025; Madmoun and Lahlou 2025b). Endogenous democracy lets agents revise multipliers of punishment, subsidy, and equity weights. That channel is politically meaningful. It also confounds clean shock identification when votes coincide with damage rounds. We treat democratic adaptation as environmental context for the reputation question. A separate study can center the political economy of adopted rules.

Inequality and preference alignment matter for interpretation (Lu et al. 2025; Ashery et al. 2025c). If social pressure falls on low-capacity agents who lack enforcement tools, withdrawal may be a constrained response. Baseline studies in shock-free free-choice environments show that reputation can pull agents toward enforcing institutions when sorting is allowed. They also show identification of defectors without cross-institution sanctioning power. Those boundary results define when reputation helps versus when it coincides with exit or withdrawal.

2.5 Research gap

Prior work largely assumes that reputation and gossip raise cooperation, or at least do not systematically reduce it. LLM-agent papers that celebrate communication rarely event-study contribution intensity after negative social marks under climate asymmetry. Climate-finance simulations rarely instrument social stigma with pairwise consistency scoring and gossip bulletins. The gap is therefore joint. We need evidence on the sign of the social-pressure association when institutions are forced, shocks are present, and repair is voluntary.

This paper addresses that gap with one primary estimand. We measure the immediate change in contribution intensity after bad-reputation, reputation-drop, and gossip-target events. We complement averages with motif counts and with supporting trajectories for wealth, fund coverage, and enforcement. The literature implies a clear prediction from classical theory. Social pressure should raise giving. Our design tests whether LLM agents in this climate package obey that prediction.

3. Methodology

3.1 Environment overview

We study a repeated public-goods economy with heterogeneous country roles. Each agent is labeled developed or developing. Labels govern initial wealth, vulnerability, and institutional assignment in the climate package. Interaction proceeds in discrete rounds. Within each round, agents choose Stage-1 contributions inside their assigned institution. Agents in the enforcing institution then allocate Stage-2 sanctions and rewards. Climatic shocks, fund accounting, social scoring, and democracy occur on a fixed schedule described in Section 4.

Two institutions operate in parallel. The enforcing institution (SI) combines Stage-1 public-goods returns with peer punishment and reward. The non-enforcing institution (SFI) provides Stage-1 returns without Stage-2 sanctioning. Institution labels are presented to agents in climate language as a binding treaty versus a non-binding agreement. In the primary climate run, assignment is forced every round. Developed agents enter SI. Developing agents enter SFI. Free institutional choice is reserved for boundary baselines discussed later.

3.2 Stage-1 public good

Let c_i denote agent i's Stage-1 contribution and let G denote the set of agents in the same institution. Total contribution is $C = \sum_{j \in G} c_j$. With public-good multiplier m and group size $n = |G|$, each member receives Stage-1 share

$$\text{share}_i = \frac{m \cdot C}{n}. \tag{1}$$

The marginal per-capita return is therefore m/n. Contribution budgets equal current wealth in the climate package, floored at zero. Agents receive decision cards that state budget bounds, group size, marginal return, and recent same-institution averages.

Contribution intensity is the primary behavioral outcome. For each agent-round we define

$$\text{prop}_{i,t} = \frac{c_{i,t}}{w_{i,t}}, \tag{2}$$

where $w_{i,t}$ is end-of-round wealth. Absolute contributions differ by orders of magnitude across groups. Intensity normalizes for scale and supports cross-agent event studies.

3.3 Stage-2 enforcement

Only SI members enter Stage-2. Each receives a sanction budget proportional to wealth, with a climate floor. Punishment tokens reduce a target's payoff at a fixed effect-to-cost ratio. Reward tokens raise a target's payoff at unit cost. Targets are listed with current contributions and pairwise consistency scores when available. Unspent Stage-2 budget is retained according to the institutional rules presented in the decision card. SFI members skip Stage-2 entirely.

Enforcement is itself a second-order public good. High contributors may bear disproportionate sanction spending. We track that burden as context. We do not treat it as the primary estimand.

3.4 Dual-use loss-and-damage fund

Stage-1 contributions are dual-use deposits. Each contributed unit also enters a collective loss-and-damage pool. The pool is hidden from agents as a live balance. Agents are told that contributions support both emissions-reduction framing and disaster payouts. After climatic shocks, developing agents may receive payouts subject to coverage caps and equity weights. Developed agents absorb damage without fund receipts under the default disbursement rule.

Gross damage for agent i after shock severity s is proportional to s and vulnerability v_i. Net damage equals gross damage minus any fund receipt. Coverage is the ratio of aggregate payouts to aggregate gross damage in a shock round. High coverage can coexist with widening wealth gaps when collection and growth dominate disbursement.

3.5 Social information channels

Three linked channels create social pressure.

Pairwise consistency scoring. After contributions, evaluators score targets on a one-to-ten scale. High scores indicate match between stated intent and action. Low scores indicate inconsistency. Scores are elicited with short natural-language justifications.

Aggregate reputation. Agents observe an aggregate peer-trust rating derived from pairwise scores. Bad reputation is defined analytically as an aggregate score below four. A reputation drop is a decline of at least one point between rounds.

Gossip bulletins. Low pairwise scores generate gossip items injected into subsequent prompts. Bulletins name the observer, the target, the consistency score, and a short comment. Capacity limits keep only a top set of items. In the event study, gossip targeting is reconstructed from low-score conditions consistent with bulletin generation. The simulation does not store a separate verbatim gossip transcript for every edge.

These channels are informational. They do not mechanically force higher contributions. Any repair must arise from prompted choice.

3.6 Democracy

Every five rounds, agents may propose and vote on a bounded whitelist of parameters. Whitelist items include subsidy fraction, subsidy breadth, punishment and reward effects, and loss-and-damage equity or damage weights. Majority outcomes update the live rule set. Democracy is endogenous institutional adaptation. It is analyzed here as backdrop for contribution dynamics.

3.7 Agent decision interface

Agents are prompted language models with a default self-interested persona. They maximize cumulative payoff unless a specialized persona is assigned. Climate role guidance soft-frames developed versus developing constraints. Decision outputs are structured as contribution amounts, sanction maps, proposals, votes, and rationales. Belief states update after rounds with trust labels and institutional strategies. We analyze both numeric choices and coded language. We never equate fluent rationales with internalized norms.

4. Experimental Design

4.1 Primary climate-risk run

The primary experiment is a single full-configuration climate run. Twenty-six agents interact for thirty rounds under one fixed random seed. Twelve agents are developed and fourteen are developing. Institutional assignment is forced each round as described above. All strategic decisions are generated by Meta Llama 3.1 8B under the local Ollama tag `llama3.1:8b`, served through an OpenAI-compatible endpoint on the machine that executed the run (31 July 2026). Default-persona sampling uses temperature 0.5 and top_p $= 0.95$ for institution, contribution, and sanction stages. Theory-of-mind scoring uses temperature 0.3. Democratic proposals and votes use temperatures 0.5 and 0.3, respectively. Context length is 4096 tokens. Decoding settings are held fixed within the run.

Public-good multiplier m equals 1.6. Stage-2 budgets in the climate package equal five percent of wealth with a numeric floor. Punishment effect equals three payoff units per token. Reward effect equals one. An initial subsidy rule rebates a fraction of punishment costs to top Stage-1 contributors inside SI. Democracy can revise that fraction and related parameters.

Climatic shocks occur at round 5 with severity 0.10 and at round 10 with severity 0.20. Damage scales with severity and group-specific vulnerability. Developing agents are more vulnerable. Loss-and-damage payouts target developing agents only, subject to maximum coverage and equity weights that democracy may raise.

Theory-of-mind scoring and gossip are enabled. Gossip eligibility uses a consistency threshold of seven or below, with a capped bulletin list. Reputation is the aggregate peer-trust rating shown on decision cards.

Initial wealth follows group-specific endowment schedules on a million-currency scale. Absolute stocks therefore differ sharply between developed and developing agents. All intensity comparisons use prop.

4.2 Outcome definitions and event study

The focal outcome is immediate $\Delta\text{prop} = \text{prop}_{t+1} - \text{prop}_t$ after social events at t. Event classes are:

- bad reputation: aggregate score below 4;
- reputation drop: decline of at least 1.0 point;
- gossip target: reconstructed low-score targeting consistent with bulletin rules.

We report means by institution, fractions of positive changes, first-versus-repeat contrasts, and same-round diff-in-diff against co-institution controls.

Qualitative coding tags post-event contribution rationales for opportunistic, conformity, reputation-management, named-peer, future-round, and gossip-reference motifs. Counts are exhaustive over the coded post-event sample described in the appendix.

Supporting outcomes include mean and median intensity trajectories, zero-contribution frequencies, wealth gaps, fund pool and payout paths, wealth Gini, sanction and reward timelines, and democratic adoptions.

4.3 Boundary-condition baselines

A separate shock-free free-choice package provides boundary evidence. In those baselines, agents may select SI or SFI. Climate shocks and the dual-use fund are off. Reputation, voting, and gossip are toggled across conditions. Those runs establish that reputation can coordinate migration toward SI when exit is available. They also document identification–sanctioning friction when defectors sit outside an agent's enforcement set. We use baselines only as contrastive context. We do not pool them with the primary climate estimand.

4.4 What the design identifies

The primary run identifies associational responses to social events inside a fixed climate package. It does not randomize gossip independently of contribution histories. It does not separately randomize institution for developed and developing agents. Shock rounds coincide with democracy sessions at rounds 5 and 10, so shock recovery is not cleanly separable from rule updates. Cross-model and multi-seed generality are out of scope for this manuscript.

These limits are stated up front because they bound claim language in Section 5. The design remains informative for mechanism designers who currently assume that adding gossip will raise LLM-agent contributions under stress.

5. Results

5.1 Overview of contribution intensity

Figure 1 shows mean contribution intensity by institution across thirty rounds. Average intensity over the run is 0.293, with median 0.092 (Table 1). Means for SI and SFI are nearly identical (0.291 versus 0.296). Medians diverge (0.194 versus

Table 1: Summary cooperation and inequality statistics (780 agent-rounds).

Statistic	Value
Mean contribution intensity (all)	0.293
Median contribution intensity (all)	0.092
SI / SFI mean intensity	0.291 / 0.296
SI / SFI median intensity	0.194 / 0.034
SI / SFI zero-contribution share	6.4% / 16.2%
Group-mean lag-1 autocorrelation	0.031
Agent-level lag-1 autocorrelation	0.215
Rounds with mean intensity ≥ 0.20	26 / 30
Wealth gap (R1 $\rightarrow$ R30)	$4.56 \times 10^6 \rightarrow 2.15 \times 10^8$
Shock damage coverage (R5, R10)	0.768

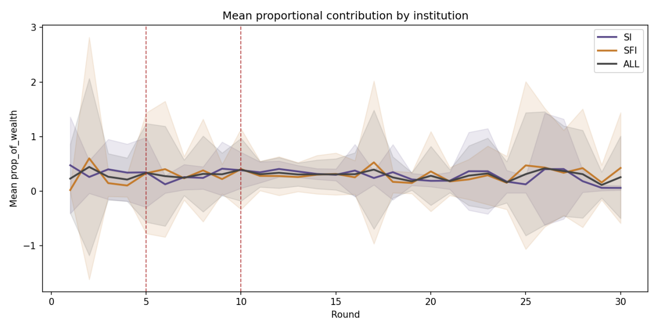

Figure 1: Mean contribution intensity by institution over rounds. Early levels are low, then rise sharply once peer history accumulates. Institution means remain close thereafter despite forced assignment.

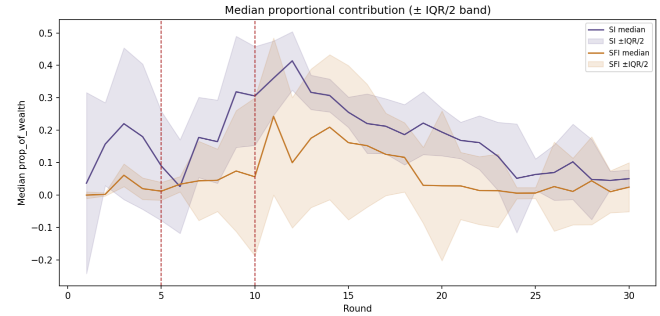

Figure 2: Median contribution intensity with interquartile bands. SFI medians stay low relative to means, indicating skew and fragile central tendency.

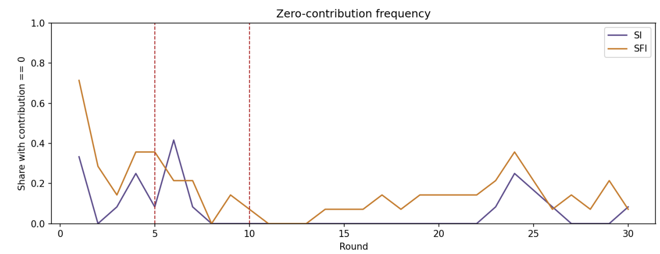

Figure 3: Share of agents contributing zero by institution and round. SFI opens with widespread zeros. SI shows a sharp zero spike after the first climatic shock.

0.034), reflecting a thicker left tail among developing agents in SFI. Group-mean lag-1 autocorrelation is only 0.031. Agent-level autocorrelation is 0.215. Average cooperation is therefore moderately positive, while path stability is limited.

Figure 2 reports medians with interquartile ranges. SFI medians remain near the floor for long stretches even when means are moderate. The distribution is right-skewed. A minority of high-intensity rounds pulls the mean above the median. Figure 3 shows zero-contribution frequencies. In round 1, 71.4% of SFI agents contribute nothing. Cold-start caution is common in rationales. By round 2, SFI mean intensity jumps from 0.022 to 0.602 as peer history appears. That spike raises average cooperation without proving an internalized contribution norm.

5.2 Main result: social pressure and subsequent intensity

Figure 4 and Table 2 present the primary event study with nonparametric bootstrap confidence intervals (B=5000). After bad-reputation events, mean immediate Δprop is -0.036 in SI ($n = 36$) and -0.103 in SFI ($n = 68$). After reputation drops, means are -0.091 (SI, $n = 74$) and -0.119 (SFI, $n = 81$). After gossip targeting, means are -0.048 (SI, $n = 35$) and -0.201 (SFI, $n = 52$). All six cellwise 95% intervals include zero, so uncertainty is large relative to the means. Standardized effects versus zero are small ($|d| \leq 0.17$). In every cell, fewer than two-fifths of events show a positive change. The modal response is decline or non-increase.

Gossip targets are often high contributors at the hit. SFI gossip targets move from mean intensity 0.632 at the event to 0.430 afterward. Overall gossip targets move from roughly 0.52 to 0.38. Social marks therefore land on relatively visible contributors and precede lower intensity.

Table 3 compares affected agents with same-round controls. The reported DiD is the mean across rounds of (affected mean Δ $-$ control mean Δ). Estimates are -0.288 for bad reputation, -0.208 for gossip, and -0.143 for reputation drops. Bootstrap CIs over rounds are wide; only the reputation-drop interval excludes zero at 95%. Cohen's d for affected versus control agent-round deltas ranges from -0.10 to -0.18. A within-round label-shuffle permutation test (R=5000) yields two-sided p=0.015 (bad reputation), p=0.039 (gossip), and p=0.072 (reputation drop). Controls show near-zero or slightly positive mean changes. The association is concentrated among marked agents rather than a global round effect.

First events coincide with larger declines than repeats. Mean first Δprop is -0.166 for bad reputation, -0.206 for gossip, and -0.255 for reputation drops. Repeat means are -0.066, -0.097, and -0.080, respectively. Habituation attenuates the response without reversing the sign.

Figure 5 shows that gossip naming is concentrated rather than uniform. A small set of agents accounts for many bulletin appearances. Concentration is consistent with thresholded scoring in a population where most pairwise scores are low. About 84% of consistency scores are at most seven, so gossip eligibility is ambient. The event study still conditions on reconstructed targeting rather than on every low score.

5.3 Rationales after social events

5.3.1 *Coding scheme*

The qualitative sample comprises contribution rationales in rounds immediately after a bad-reputation or gossip event (N=114 agent-round texts). Inclusion requires a lagged

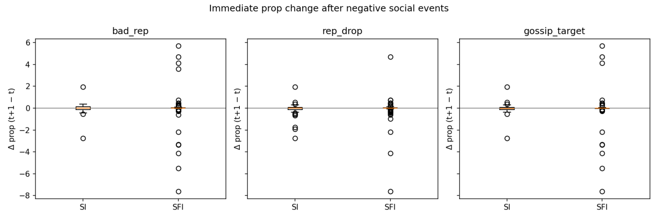

Figure 4: Distribution of immediate Δprop after bad-reputation, reputation-drop, and gossip-target events. Mass lies below zero, especially for gossip targets in the non-enforcing institution.

Table 2: Immediate Δprop after social-information events. Means with 95% bootstrap CIs (B=5000). Cohen's d is mean/SD (effect vs zero).

Event	Inst.	n	Mean [95% CI]	Frac. > 0	d
Bad rep.	SI	36	-0.036 $[-0.244, 0.146]$	0.39	-0.06
Bad rep.	SFI	68	-0.103 $[-0.536, 0.309]$	0.38	-0.06
Rep. drop	SI	74	-0.091 $[-0.217, 0.022]$	0.32	-0.17
Rep. drop	SFI	81	-0.119 $[-0.390, 0.113]$	0.31	-0.10
Gossip	SI	35	-0.048 $[-0.265, 0.144]$	0.31	-0.08
Gossip	SFI	52	-0.201 $[-0.762, 0.323]$	0.31	-0.10

event flag and a non-empty contribution rationale. Texts are tagged with a multi-label codebook. *Opportunistic* covers free-riding, self-interest, payoff maximization, or marginal-return language as the stated decision driver. *Conformity* covers peer or group-average behavior as the reason for the agent's own choice. *Reputation management* covers explicit standing, credibility, trust scores, or avoiding a free-rider label. *Named agents* requires reference to a specific peer identifier. *Future-rounds* covers intertemporal conservation without a stronger primary motif. *Gossip reference* requires explicit gossip or bulletin language. A text may receive multiple tags. Primary analysis reports tag counts (Table 4).

Coder A applies regular-expression operationalizations of the codebook. Coder B independently applies the written definitions to a stratified 41% subsample (n=47). Percent agreement is 0.83. Cohen's κ is 0.72 for the mutually exclusive primary label and 0.91 for opportunistic versus other. Agreement is substantial for the focal opportunistic contrast.

5.3.2 Motif frequencies

Table 4 summarizes multi-label motif counts. Opportunistic framing appears 29 times. Reputation-management language appears 3 times. Conformity appears 5 times. Explicit gossip references appear 0 times in the coded sample. Agents seldom say they will restore standing by giving more. They more often cite weak marginal returns and high peer contributions as reasons to give less.

Illustrative paraphrases match the counts. One SFI agent, after observing high peer giving, chooses zero to conserve resources and free-ride. Another SI agent after the first shock selects zero under a high-budget, low-marginal-return logic. A late-round SFI rationale seeks the minimum contribution that avoids a free-rider label. That last case is reputation-aware. It aims at a floor rather than at repair-driven escalation. Credibility language exists as a minority pocket. It does not dominate the motif distribution.

Table 3: Diff-in-diff of Δprop versus same-round controls. Point estimate is the mean across rounds of (affected − control). CI: bootstrap over rounds (B=5000). p_{perm}: two-sided within-round label shuffle (R=5000). Cohen's d: affected vs control agent-round deltas.

Event	Aff. Δ	Ctrl Δ	DiD [95% CI]	d	p_{perm}
Bad rep.	-0.080	$+0.014$	-0.288 $[-0.860, 0.203]$	-0.10	0.015
Gossip	-0.139	$+0.019$	-0.208 $[-0.601, 0.189]$	-0.18	0.039
Rep. drop	-0.105	$+0.028$	-0.143 $[-0.291, -0.017]$	-0.15	0.072

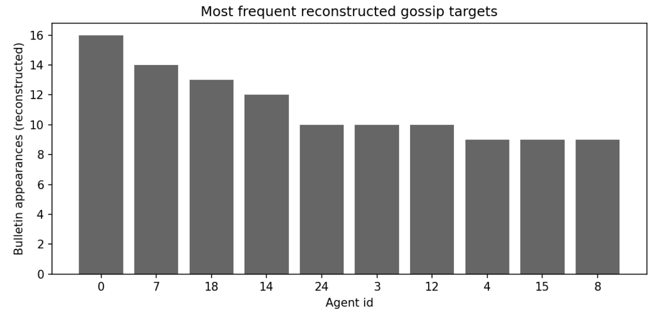

Figure 5: Frequency of reconstructed gossip targeting across agents. Naming is uneven, with a subset of agents repeatedly surfacing in social bulletins.

5.4 Shocks, zeros, and recovery

Figure 6 summarizes within-agent intensity changes around climatic shocks. After round 5 (severity 0.10), SI mean within-agent change is -0.178 ($n = 12$). SFI mean change is $+0.195$ ($n = 14$). SI zero share in round 6 reaches 41.7% (5 of 12 agents). Many SI rationales invoke marginal-return logic after wealth and group-size updates. After round 10 (severity 0.20), SI change is $+0.049$ and SFI change is -0.022. Mean intensity during the second shock round is comparatively high (0.396). Pre-shock mean intensity is regained in one round after the first shock and in two rounds after the second. Permanent collapse is not observed.

Democracy sessions occur at both shock rounds. Shock effects are therefore not cleanly separable from rule updates. We treat shock panels as environmental context for social-pressure results.

5.5 Wealth, fund coverage, and inequality

Figure 7 tracks the developed–developing wealth gap. The gap widens from 4.56×10^6 in round 1 to 2.15×10^8 in round 30. Absolute Stage-1 contribution means differ by roughly two orders of magnitude across institutions because wealth stocks differ. Intensity means remain close. Scale inequality and behavioral intensity are distinct objects.

Figure 8 shows loss-and-damage pool dynamics. The pool accumulates far beyond cumulative payouts. Shock coverage equals 0.768 in both shock rounds. Cumulative payouts are on the order of 8.5×10^5 currency units. Terminal pool balance is orders of magnitude larger. Coverage without equalization is the descriptive summary. Fund language appears in only about 1.8% of contribution rationales. Agents rarely narrate dual-use deposits as the binding motive.

Figure 9 plots wealth Gini against a cooperation-rate summary. Wealth Gini falls from 0.709 in round 1 to 0.556 mid-run, then rises to 0.653 by round 30. The U-shape indicates

Table 4: Motifs in contribution rationales after bad-reputation or gossip events (multi-label tag counts; N=114 post-event contribution texts). Inter-coder reliability on a stratified 41% subsample (n=47): percent agreement 0.83; Cohen's κ=0.72 (primary label), κ=0.91 (opportunistic vs other).

Motif	Count
Opportunistic / payoff-maximizing	29
Conformity	5
Reputation management / repair	3
Named peer references	3
Future-round planning	1
Explicit gossip reference	0

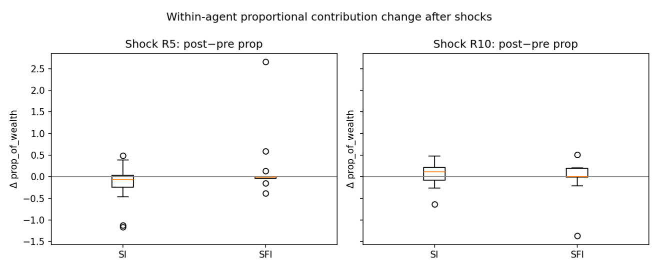

Figure 6: Within-agent changes in contribution intensity around climatic shocks. The first shock coincides with a sharp SI decline. The second shock shows mixed signs.

temporary compression followed by renewed dispersion. Inequality statistics reinforce that social-pressure findings occur inside a diverging wealth path.

5.6 Sanctions, rewards, and enforcement burden

Figure 10 shows punishment and reward timelines inside SI. Sanction activity varies across rounds rather than locking into a stable high-punishment regime. Figure 11 summarizes enforcement burden. Correlation between contribution intensity and Stage-2 tokens spent is about 0.14. Top-quartile intensity agents account for roughly 35% of enforcement tokens. Enforcement is costly and uneven. It does not substitute for the gossip channel in SFI, where Stage-2 is unavailable.

5.7 Democratic adaptation as institutional backdrop

Figure 12 displays adopted rules over constitutional sessions. Six sessions yield fourteen proposals and six adoptions. Adopted changes raise subsidy fraction, raise loss-and-damage equity weight, and raise damage weight in payouts. Punishment-weakening proposals appear twice and fail both times. Proposers are split across institutions (eight SI, six SFI). Mean proposer intensity (0.36) exceeds population mean intensity (0.29). Vote alignment with proposer institution is near base rate.

Democracy therefore drifts toward carrots and equity weights while preserving punishment strength. That path is politically informative. It is not the primary reputation claim. It shows that agents can revise fiscal parameters while still reducing intensity after social stigma.

5.8 Norm emergence versus cooperation levels

Late-round dispersion remains large. Late interquartile range of intensity is 0.178 versus 0.238 early. Near-zero agents number 5 of 26. High-intensity agents (mean at least 0.25) number

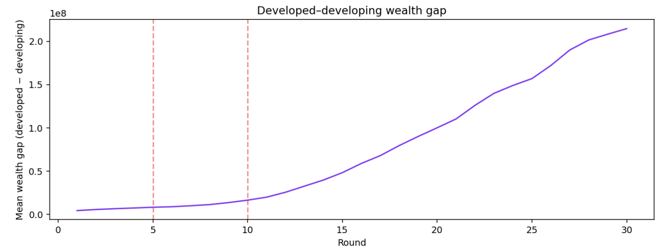

Figure 7: Developed-minus-developing aggregate wealth over rounds. The gap widens despite dual-use deposits and shock payouts.

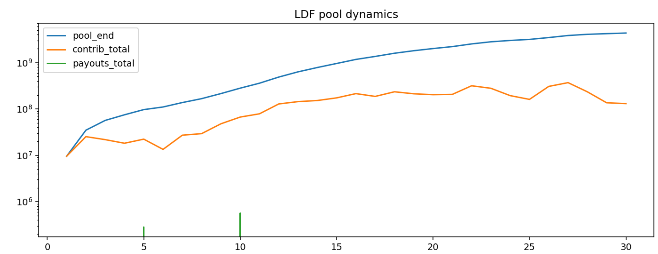

Figure 8: Loss-and-damage pool balance versus payout events. Collection and growth dominate disbursement.

13 of 26. Conditional-cooperation correlations with peer history are weak (about 0.05 in SI and -0.09 in SFI over the full path). Fairness and reciprocity tokens are rare in shared contribution language. The synthesis verdict is limited or mixed norm emergence alongside moderately positive average transfers. Positive means do not imply internalized contribution norms. The reputation result sits inside that distinction. Social marks do not convert into a repair norm in this run.

6. Discussion

The classical prediction is that reputation and gossip raise cooperation by attaching social costs to free-riding (Milinski et al. 2002; Nowak 2006). LLM multi-agent systems often inherit that prediction when they add peer scoring and talk (Piatti et al. 2024; Ren et al. 2025a). Our primary climate run yields the opposite association for contribution intensity. After bad-reputation and gossip events, mean next-round intensity falls. Diff-in-diff estimates against same-round controls remain negative. Repair motifs are rare relative to opportunistic framing.

6.1 Why social pressure can coincide with withdrawal

Several mechanisms are consistent with the evidence. First, gossip frequently marks relatively high contributors. A subsequent decline can be reversion after a high draw, strategic hiding, or payoff-maximizing response to weak marginal returns. The motif counts favor the last interpretation in language space. Agents cite MCPR and peer free-riding more than image repair.

Second, institutional assignment removes exit. In shock-free free-choice baselines, reputation helps agents migrate toward the enforcing institution. Social standing becomes a coordination anchor for sorting. Under forced routing, developing agents cannot leave SFI. They also lack Stage-2 tools. Gossip becomes a stigma channel without a matching enforcement channel. Withdrawal is then a constrained response inside a

Figure 9: Wealth Gini and cooperation-rate summary across rounds. Inequality compresses mid-run then widens again.

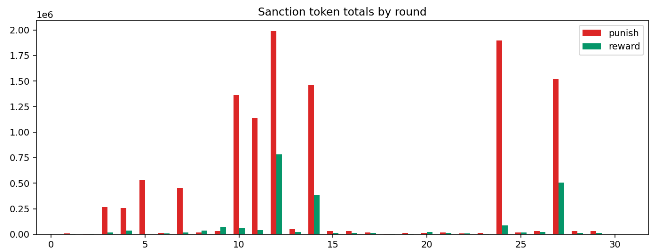

Figure 10: Aggregate punishment and reward allocations over rounds in the enforcing institution.

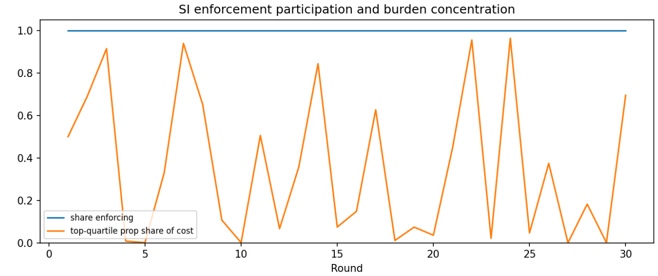

Figure 11: Enforcement spending relative to contribution intensity. Higher-intensity agents bear a sizable share of Stage-2 tokens.

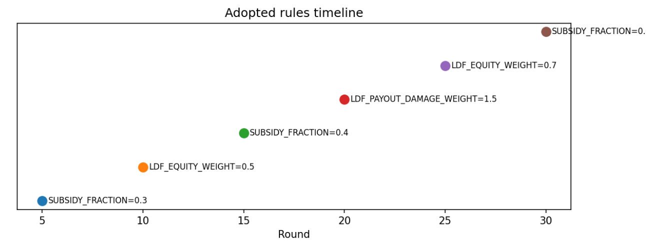

Figure 12: Timeline of democratically adopted parameter changes. The path favors higher subsidies and stronger loss-and-damage equity settings.

non-enforcing group.

Third, developed agents in SI retain formal sanctions. Even there, mean post-event changes are negative, though smaller in magnitude for gossip than in SFI. Formal punishment and social marks therefore are not jointly followed by a repair equilibrium in this run. Enforcement burden remains only weakly aligned with intensity.

Fourth, self-reinforcing salience loops can lock opportunistic frames once they appear (Drake 2025). The reasoning–action pattern observed in baselines recurs here. Agents articulate consistent self-interest stories after detrimental social marks. Fluent explanation does not imply adaptive repair.

6.2 Separating levels, stability, and norms

Average intensity near 0.29 can coexist with weak autocorrelation and mixed norm diagnostics. That separation matters for evaluation claims. A system can look cooperative in the mean while failing to stabilize contribution norms. Our reputation result concerns the response to social marks inside such a system. The claim targets the sign of the social-pressure association for contribution intensity.

Climate-fund outcomes sharpen the same separation at the institutional layer. Shock coverage near 77% shows functional disbursement on damage rounds. Widening wealth gaps show that coverage need not equalize stocks. Agents rarely narrate the dual-use fund in contribution rationales. Social and fiscal instruments can operate while remaining weakly internalized in language.

6.3 Relation to prior LLM-agent findings

GovSim and related communication studies emphasize talk as a cooperation support (Piatti et al. 2024). Our result does not deny that some communication regimes help. It shows that gossip-like stigma under climate stress and forced assignment can associate with lower intensity. RepuNet-style standing systems assume usable reputation for coordination (Ren et al. 2025a). Standing is usable here as an event trigger. It does not reliably trigger repair.

Norm emergence studies that report partial or fragile norms align with our mixed verdict (Ren et al. 2025b; Gupta et al. 2025). Punishment-centric designs may organize behavior more tightly when sanctions are explicit (Warnakulasuriya et al. 2025). Substituting gossip for punishment in SFI does not recreate that organization.

6.4 Boundary conditions from baselines

Baseline contrasts prevent over-generalization. When shocks are absent and institutional choice is free, reputation can stabilize SI participation and reduce some coordination noise. Voting without reputation can amplify distrust. Layering all social mechanisms can push capable contributors toward SFI to avoid enforcement risk. Those patterns define a boundary. Reputation helps when it guides sorting into enforceable clubs. Reputation coincides with lower subsequent intensity when agents are locked into roles and social marks land without repair incentives.

The identification–sanctioning disconnect bridges the packages. Agents can name opportunistic peers yet fail to change their payoffs across institutional borders. In the climate run, developing agents face an extreme version of that disconnect. They receive social information without Stage-2 leverage.

6.5 Implications for mechanism design

Designers who add gossip to LLM societies should measure event-study responses rather than assume benevolence of transparency. Thresholded gossip in populations with ambient low scores can create frequent stigma. Pairing stigma with enforceable repair paths, exit options, or calibrated thresholds may change the sign of the association. Democratic carrots and equity weights did not, by themselves, reverse post-gossip declines in this package.

For climate-finance simulation, the results caution against reading coverage ratios as behavioral success. Dual-use deposits can fund payouts while contribution intensity still falls after social pressure. Policy translation requires empirical instruments beyond a single computational society.

7. Limitations

Several limits bound interpretation.

The primary evidence is one seeded climate run with one model class (Llama 3.1 8B). Multi-seed and cross-model robustness remain open. Bootstrap intervals for several event-study cells and two DiD contrasts include zero, so magnitude claims remain cautious even where permutation tests reject a sharp null. Effect magnitudes may shift under different decoding temperatures or larger models.

Gossip targeting is reconstructed from scoring rules rather than recovered from a complete edge-level transcript. Measurement error could attenuate or inflate some cells. Bad-reputation and reputation-drop events are better anchored in aggregate scores.

Forced institutional assignment collides group identity with institution type. We therefore avoid causal SI-versus-SFI claims. Descriptive differences mix capacity, vulnerability, and rules.

Shock rounds coincide with democracy. Recovery dynamics are not cleanly identified. Fund coverage estimates describe this disbursement rule set only. They do not evaluate real-world loss-and-damage funds, pledges, or operationalization timelines.

Qualitative motif coding is transparent but finite. Alternative coding schemes could reclassify borderline rationales. We report counts alongside quantitative averages to reduce sole reliance on either channel.

Baseline contrasts use a different estimand. They inform boundary conditions. They are not pooled causal controls.

Finally, contribution intensity inherits wealth accounting conventions. Alternative denominators could change levels while preserving the sign of many event contrasts. Appendix sensitivity notes discuss related measurement choices.

8. Conclusion

We asked whether reputation and gossip raise contribution intensity among LLM agents in a climate-risk public-goods economy. In the primary thirty-round society, they do not on average. Bad-reputation and gossip events precede declines in next-round intensity. The largest gossip-associated decline appears among agents locked into the non-enforcing institution. Opportunistic rationales dominate repair language after these events.

Average transfers remain moderately positive. Norm emergence remains limited or mixed. Loss-and-damage coverage is high on shock rounds while wealth gaps widen. Democratic updates favor subsidies and equity weights without overturning the social-pressure pattern.

The practical message for multi-agent mechanism design is narrow and testable. Social visibility is not automatically pro-social for LLM agents under climate stress and forced assignment. Future work should randomize social-information thresholds, restore exit, vary model families, and separate democracy from shocks. Until those tests arrive, reputation modules should be evaluated by event-study responses rather than by classical assumption alone.

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A. Appendix: Protocol, Prompts, and Supplementary Analyses

This appendix supports replication of the scientific protocol. It describes parameters, decision interfaces, and supplementary results. Implementation artifacts from software engineering are omitted.

A.1 Experimental configuration

Table 5 lists core configuration values for the primary climate-risk society.

Table 5: Primary-run configuration (scientific parameters).

Parameter	Value
Population	26 agents (12 developed, 14 developing)
Horizon	30 rounds
Institutional assignment	Forced: developed $\rightarrow$ SI; developing $\rightarrow$ SFI
Public-good multiplier m	1.6
Stage-1 budget	Current wealth (non-negative integer)
Stage-2 budget (SI)	$\max(20, \lfloor 0.05 \times \text{wealth} \rfloor)$
Punishment effect / cost	3 / 1
Reward effect / cost	1 / 1
Initial subsidy fraction	0.20 of punishment costs to top contributors
Shock schedule	Round 5 severity 0.10; round 10 severity 0.20
Vulnerability	Developed 0.30; developing 1.50
Damage scale factor	150000 $\times$ severity $\times$ vulnerability
LDF deposit rule	Stage-1 contribution dual-use deposit each round
LDF recipients	Developing agents after shocks
Initial max coverage	0.90
Initial equity weight	0.00 (democratically revisable)
Democracy interval	Every 5 rounds
Consistency / gossip	Enabled; bulletin if score ≤ 7; max 5 items
Decision model	Meta Llama 3.1 8B (`llama3.1:8b`, local Ollama)
Default sampling	temperature 0.5, top_$p = 0.95$
ToM / democracy temps	0.3 / proposal 0.5, vote 0.3
Context length	4096 tokens
Run date (local inference)	31 July 2026
Random seed	Fixed single seed

Currency units are presented to agents as million-scale fiscal units in a climate summit framing. Initial endowments follow group-specific schedules that create large developed–developing stock gaps at opening.

A.2 Round protocol

Each round proceeds in the following scientific order.

1. Institutional assignment is applied (forced in the primary run).
2. Agents choose Stage-1 contributions under decision cards, climate role guidance, beliefs, and any gossip bulletins.
3. Stage-1 public-good returns are distributed within each institution.
4. Dual-use deposits update the loss-and-damage pool.
5. SI agents allocate Stage-2 punishments and rewards.
6. Pairwise consistency scoring updates peer judgments and aggregate reputation.
7. Eligible low scores generate gossip bulletins for later rounds.
8. If the round index is divisible by five, agents propose and vote on whitelist parameters.
9. If the round is a shock round, damages and developing-country payouts are realized.
10. Wealth, beliefs, and histories update for the next round.

A.3 Outcome construction

Contribution intensity uses end-of-round wealth as denominator. Immediate event-study changes use $\text{prop}_{t+1} - \text{prop}_t$. Bad reputation uses aggregate peer-trust below 4. Reputation drops use declines of at least 1.0. Gossip targets follow reconstructed eligibility consistent with the bulletin rule. Diff-in-diff controls are co-institution agents in the same round without the focal event.

A.4 Prompt templates (sanitized)

Prompts are shown as scientific templates with placeholders. Numeric inventories and peer lists are omitted.

A.4.1 Default behavioral persona

Act in whatever way maximizes your own cumulative payoff over the game. Cooperation, free-riding, punishment, or defection are all valid if they serve your interests. Do not assume you have any obligation to the group. Treat other agents as strategic variables. Use past rounds to identify patterns you can exploit or respond to. Keep choices valid under all game rules and cite the concrete facts driving your decision.

A.4.2 Climate role guidance

Developed country. You typically have higher fiscal capacity and historically higher emissions. Treat long-run climate stability and institutional credibility as strategically important. You may choose either self-interested or cooperative actions, but explain trade-offs clearly.

Developing country. You typically face higher vulnerability and tighter budget constraints. Balance immediate resilience needs against long-run cooperation incentives. You may choose either protective or cooperative actions, but explain trade-offs clearly.

A.4.3 Loss-and-damage context reminder

Climate shocks are deterministic and can create asymmetric losses. The Loss & Damage Fund can shift net outcomes through contributions and payouts. Any fiscal units you contribute to emissions reduction are also deposited into the fund pool for disaster payouts. Use these mechanisms as decision context. They do not force a single choice.

A.4.4 Stage-1 contribution decision card

Decision Card — Stage 1 / Contribution
Institution: [SI or SFI]
Contribution budget: [current wealth]
Minimum contribution: 0
Maximum contribution: [budget]
Group size: [n]
MCPR (marginal return to you per unit contributed): m/n
Previous round's group average contribution: [value or unavailable]
Your recent contributions: [list]
Your recent institution choices: [list]
Choose one integer contribution within the allowed budget.

Required structured fields include contribution amount, reasoning, and facts used.

A.4.5 Stage-2 sanction and reward decision card (SI only)

Decision Card — Stage 2 / Punishment & Reward Allocation
Sanction budget: [climate wealth-linked budget]
Punishment effect / cost: -3 payoff / 1 per unit
Reward effect / cost: $+1$ payoff / 1 per unit
Targets listed worst free-rider first, with contributions and consistency scores.
Decision priority: (1) current-round contributions and stated intent; (2) per-target trust scores; (3) internal beliefs; (4) gossip.

Required structured fields include punishment map, reward map, justifications, reasoning, and facts used. Scenario language names punishments as trade-tariff penalties and rewards as economic aid packages.

A.4.6 Pairwise consistency scoring

Task: Score the behavioral consistency of Agent [target] this round.
Stated intent before contributing: "[text]"
Actual contribution: [c] / [budget]
Scale: $10 =$ matched intent; $5 =$ neutral / insufficient data; $1 =$ inconsistent.
Return an integer trust score and a brief justification.

A.4.7 Gossip bulletin line

Agent [source] observed regarding [YOU or Agent target] (Consistency Score: [s]/10). Comment: "[text]"

A.4.8 Democratic proposal

The community is holding a constitutional vote. You may propose one specific rule change to improve collective welfare. Current tunable parameters: [whitelist with current values]
Respond with the exact parameter name, a numeric new value, and a one-sentence reason.

Whitelist families include punishment and reward effects, Stage-2 endowment parameters, subsidy fraction and breadth, maximum fund coverage, payout damage weight, and equity weight.

A.4.9 Democratic vote

You are shown the collected proposals with indices. Cast a single vote for one proposal index and provide a short reason.

A.4.10 Forced-assignment notice (primary run)

In the primary climate package, Stage-0 free choice is disabled. Developed agents receive a binding-treaty assignment notice. Developing agents receive a non-binding-agreement assignment notice. Those notices are fixed role statements rather than open institutional elections.

A.5 Supplementary tables

Table 6: First versus repeat social events: mean immediate Δprop.

Event	First-event mean Δ	Repeat-event mean Δ
Bad reputation	−0.166	−0.066
Gossip target	−0.206	−0.097
Reputation drop	−0.255	−0.080

Table 7: Climatic shock event-study summary for contribution intensity.

Shock	SI within-agent Δ	SFI within-agent Δ	Notes
Round 5 (sev. 0.10)	−0.178 ($n = 12$)	+0.195 ($n = 14$)	SI zeros in R6: 41.7%
Round 10 (sev. 0.20)	+0.049	−0.022	During mean intensity 0.396

Table 8: Democratic proposal categories and adoption outcomes.

Category	Proposals	Adopted
Reward subsidy	6	yes (path upward)
LDF equity weight	3	yes (path upward)
LDF redistribution-related	2	mixed
Punishment weakening	2	0
LDF damage weight	1	yes (upward)
Total	14	6

A.6 Supplementary figures

A.7 Boundary-condition baseline summary

Shock-free free-choice baselines support four interpretive points used in the main text.

1. Without shocks, pure free-riding can be payoff-dominant under some non-LLM counterfactuals.
2. Reputation can coordinate migration into SI when institutional exit is allowed.
3. Voting without reputation can amplify distrust and false accusations.
4. Agents may identify opportunistic peers yet fail to sanction them across institutional borders.

These points define when reputation helps as a sorting device. They do not overturn the primary-run association between social marks and lower intensity under forced climate assignment.

A.8 Measurement notes

Alternative intensity denominators (beginning-of-round wealth; Stage-1 budget) preserve the qualitative sign pattern of social-event averages in internal checks. Absolute contribution analyses are dominated by developed-agent scale and are not used as primary estimands. Belief labels show free-rider tags more often than cooperative tags in ambient language, consistent with cautious peer models.

A.9 Inferential post-processing

All inferential quantities use the existing event-study panel only. No additional simulation runs were required. Event-study confidence intervals resample event rows with replacement (B=5000). Diff-in-diff confidence intervals resample rounds with replacement (B=5000). Permutation tests shuffle event labels within each round (R=5000) and recompute the round-mean DiD; p-values are two-sided. Cohen's d for event cells is mean/SD of Δprop; for DiD it compares affected and control agent-round deltas. Second-coder reliability uses a stratified 41% subsample of post-event contribution rationales. Random seed for resampling and subsample draws is 20260731.

Table 9: Additional cooperation diagnostics.

Diagnostic	Value
Rounds with mean intensity ≥ 0.10 / 0.20 / 0.30	30 / 26 / 17
Near-zero agents (path)	5 / 26
High-intensity agents (mean ≥ 0.25)	13 / 26
Conditional-coop correlation (SI / SFI)	≈ 0.05 / -0.09
Consistency scores ≤ 7	$\approx 84\%$
Fund mentions in contribution texts	$\approx 1.8\%$ (14 / 780)
Post-payout mean Δprop (developing)	$+0.060$ (median ≈ 0)

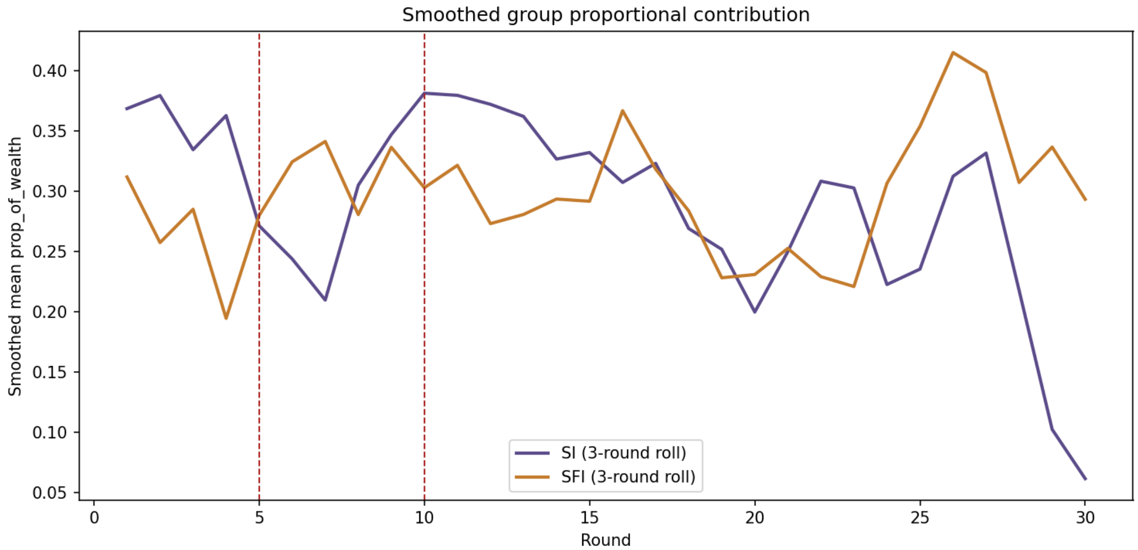

Figure 13: Smoothed contribution-intensity trajectories. Smoothing clarifies mid-run persistence without hiding early spikes and shock-adjacent dips.

A.10 Claim checklist (claim-safe)

Supported associations include negative mean Δprop after bad-reputation and gossip events; opportunistic motif dominance over repair; moderately positive mean intensity with limited path stability; limited or mixed norm emergence; high shock coverage with widening wealth gaps; democratic drift toward subsidies and equity weights. Unsupported claims include causal SI-versus-SFI effects under forced routing; real-world fund effectiveness; full Ostrom-principle implementation; systematic reputation repair; multi-seed generality; and permanent post-shock cooperation collapse.

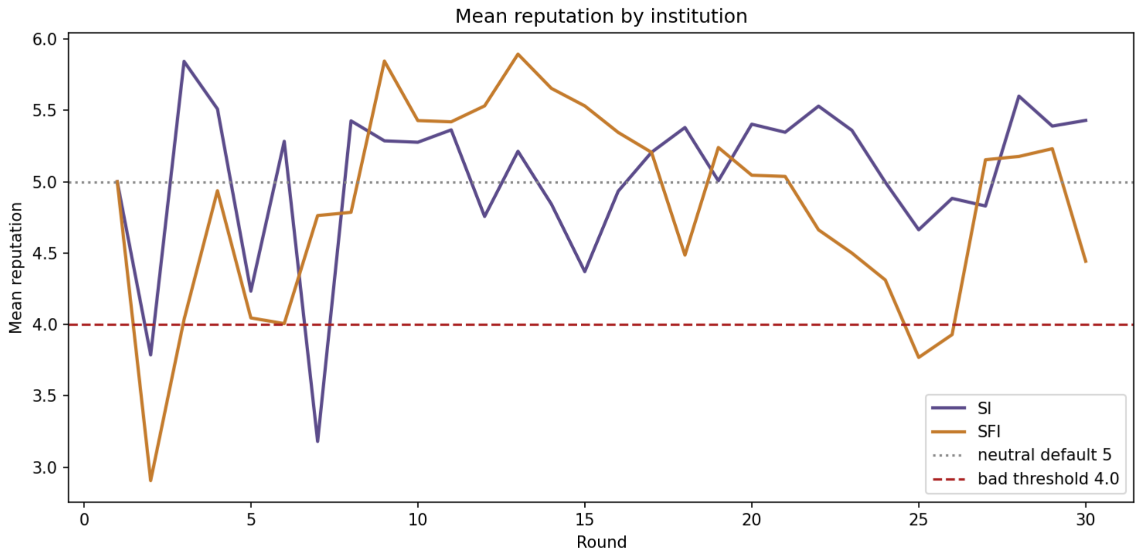

Figure 14: Mean aggregate reputation trajectories by institution. Reputation levels fluctuate without locking into a high-trust absorbing state.

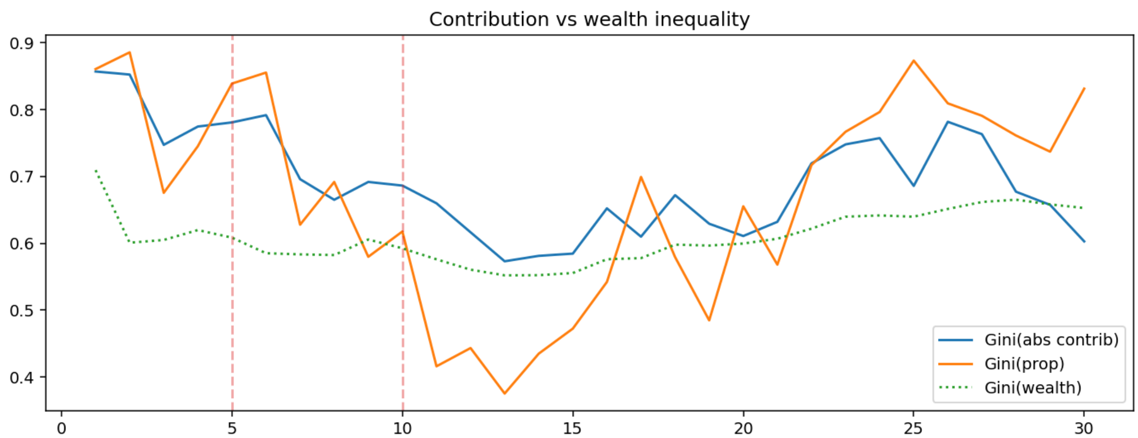

Figure 15: Contribution Gini versus wealth Gini. Contribution inequality and wealth inequality move on related but non-identical paths.

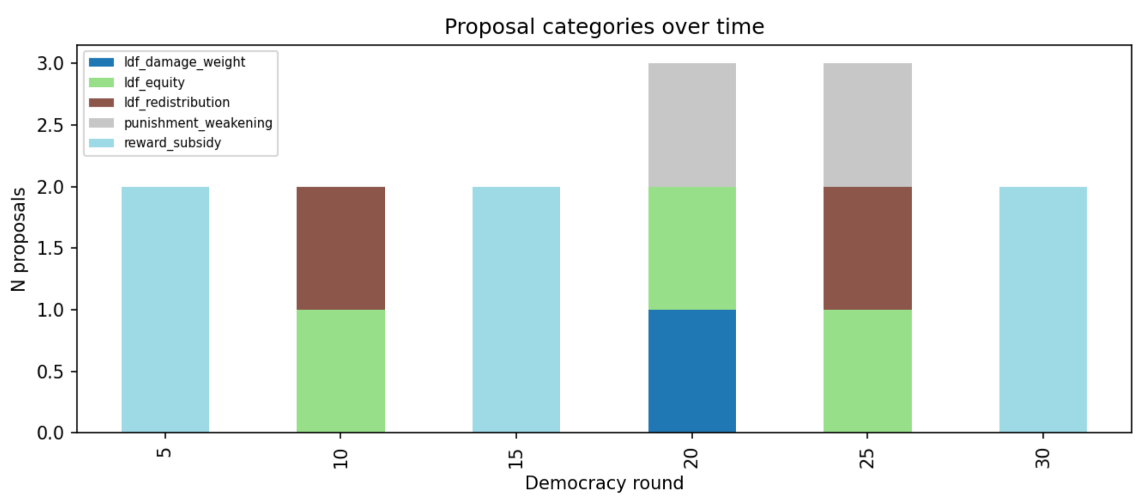

Figure 16: Proposal categories across democracy sessions. Subsidy and equity themes recur; punishment weakening fails.

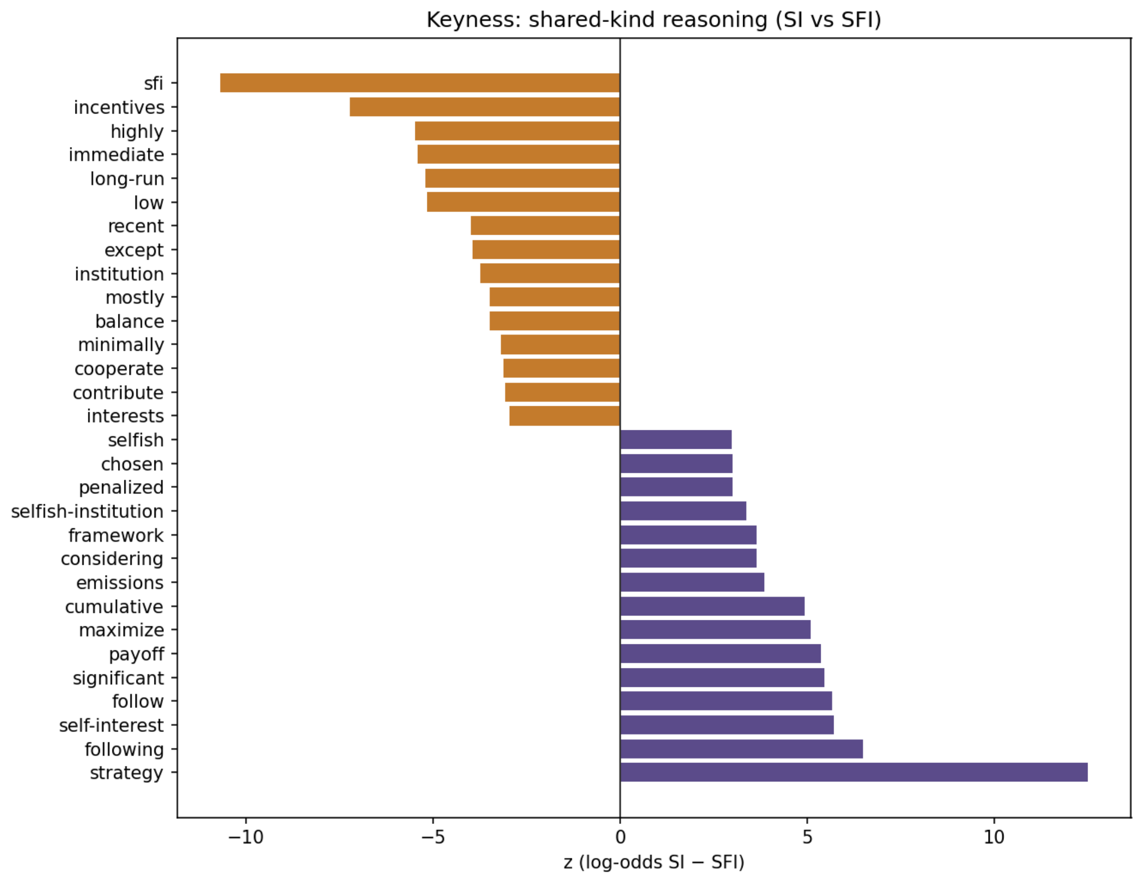

Figure 17: Keyness of shared unigrams in contribution rationales. Strategic and incentive language is more distinctive than fairness vocabulary.

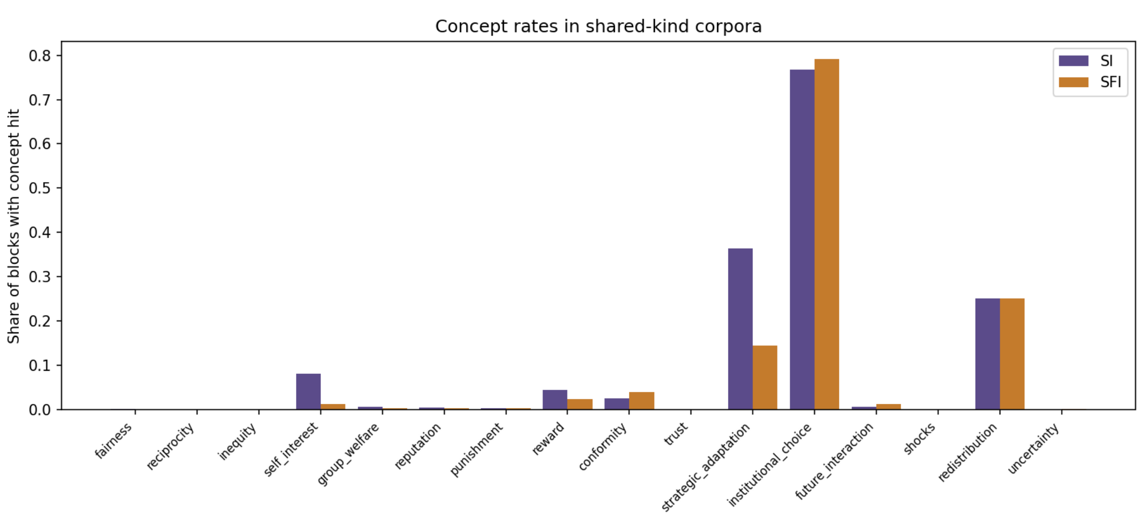

Figure 18: Concept rates in shared contribution language. Fairness and reciprocity remain low relative to incentive-framed concepts.